Kushagra Garg

Athens, GA | Kushagragarg14@gmail.com | linkedin.com/in/gargkushagra | github.com/Kushagra0514 | 470-652-0543

Education

University of Georgia Athens, GA

Bachelor of Science in Computer Science, Emphasis in Artificial Intelligence *Expected May 2028*

- Relevant Coursework: Data Structures & Algorithms, Machine Learning for Data Science, Computer Organization

Projects

Studio | *Python, FastAPI, React, Vite, Qdrant, Groq API, Docker*

- Engineered a locally-deployable, agentic Retrieval-Augmented Generation (RAG) application supporting PDF, DOCX, TXT and scraped web content, indexing 2500+ document chunks for enabling users to query an AI assistant for grounded conversational search.
- Designed a 3-route agentic query pipeline that classified requests across document retrieval, live web search, and direct conversation with 94% routing accuracy across 200+ evaluation queries.
- Built a hybrid retrieval pipeline combining dense embeddings (BAAI/bge-small-en-v1.5) with sparse BM25 scoring over a local Qdrant vector database, using Fastembed for local, privacy-preserving embedding generation resulting in improving recall from 78% to 91% representing a 16.7% overall relative improvement over dense-only retrieval.
- Optimized local Qdrant retrieval and FastEmbed inference to reduce average document-search latency from 620 ms to 340 ms, representing a 45% reduction in latency.
- Containerized with FastAPI backend, React/Vite frontend, Nginx with Docker, reducing local deployment from ∼15+ min to a single command in under 2 min and released as an open source software.

UniFlow | *Chrome Extension, D3.js, WebLLM, WebGPU*

- Developed a privacy-first Chrome extension that intercepts DegreeWorks degree-audit JSON and renders into an interactive D3.js degree map, processing 150+ academic requirements in ∼400 ms upon page loading.
- Architected a 4-stage client-side pipeline (MAIN-world interceptor, isolated-world content script, service worker, and data transformer) to capture and reshape audit data client-side with 99%+ successful audit parsing across 50 test degree audits.
- Integrated an in-browser AI advising layer via WebLLM and WebGPU, to enable 100% of student records to be processed locally, eliminating any FERPA violations and collection of personally identifiable information (PII) to external AI services.
- Generalized the extension to support any 6+ universities running DegreeWorks by pattern-matching .edu domains in Georgia.
- Proposed the project to a University of Georgia faculty member as a potential department-supported advising tool.

LocalFlow | *Python, Whisper.cpp, Groq API*

- Built a system-wide push-to-talk dictation application as an open-source alternative to subscription transcription tools, sustaining 145 wpm transcription throughput on CPU-only hardware.
- Deployed whisper.cpp base.en Q5 model entirely on CPU architecture, reducing peak memory consumption from ∼1.1 GB to ∼266 MiB, representing a 76% reduction, while maintaining 94% transcription accuracy on evaluation audio.
- Optimized local speech inference to reduce transcription latency from 1.8 s to 650 ms, achieving a 64% latency reduction without requiring any discrete GPU.
- Streamed transcripts through the Groq API (gpt-oss-20b) to correct grammar, fix capitalization, and strip filler words in real time.
- Automated clipboard injection to paste cleaned text directly into the active application, removing manual copy-paste steps.

Certifications

Machine Learning with Python Professional Certificate — Anaconda *May 2026*

Microsoft Azure AI Essentials Professional Certificate — Microsoft & LinkedIn *May 2026*

Technical Skills

Languages: Python, Java, JavaScript, C/C++
Frameworks & Libraries: FastAPI, PyTorch, Scikit-learn, TensorFlow, React, Vite, D3.js
AI/ML Tools: Groq API, OpenCode (multi-agent LLM orchestration), Ollama, LiteLLM, WebLLM/WebGPU, Qdrant
Developer Tools: Docker, Git, GitHub, Nginx